

HOEDSPRUIT, South Africa

WMO: 682910

Lat: 24.352S

Lon: 31.049E

Elev: 1719

StdP: 13.81

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	45.0	48.4	23.0	18.5	72.3	27.8	23.2	69.4	20.4	65.5	18.5	65.8	2.5	240	0.339

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.9	97.1	68.7	94.7	68.5	91.9	68.6	76.0	85.4	74.6	84.9	73.6	83.6	6.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.6	133.2	80.3	71.9	125.7	78.2	70.9	121.4	77.5	40.8	85.2	39.4	85.3	38.4	83.7	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.8	18.0	15.3		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
				DB	37.2	77.5	4.5	1.9	33.9	78.9	31.3	80.0	28.8	81.1	25.5	82.5
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.0	77.3	77.4	76.5	72.2	67.9	65.0	63.6	67.1	71.4	73.9	75.0	77.1	(6)
	DBStd	6.88	4.17	3.88	3.87	4.76	4.20	4.09	3.99	5.26	6.26	6.47	5.89	5.18	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	205	0	0	0	3	16	49	73	38	16	6	4	0	(9)
	CDD50	8031	845	768	822	666	555	451	422	529	642	740	749	842	(10)
	CDD65	2762	380	348	357	219	107	49	30	103	208	281	303	377	(11)
	CDH74	23509	3017	2638	2703	1633	953	513	393	1009	2032	2717	2634	3267	(12)
	CDH80	9359	1139	995	1089	606	285	107	76	345	895	1260	1165	1397	(13)
(14) Wind	WSAvg	6.4	7.2	7.1	6.1	5.6	4.9	5.3	5.8	6.3	7.5	7.8	6.9	6.7	(14)
(15) Precipitation	PrecAvg	24.6	4.6	3.7	3.5	1.6	0.4	0.2	0.2	0.3	0.6	1.5	3.4	4.4	(15)
	PrecMax	50.9	10.3	17.9	9.4	6.4	2.7	1.0	0.8	1.1	2.9	3.0	8.5	8.4	(16)
	PrecMin	11.9	1.1	0.8	1.2	0.1	0.0	0.0	0.0	0.0	0.0	0.3	0.7	1.3	(17)
	PrecStd	8.3	2.6	3.9	2.2	1.3	0.5	0.2	0.2	0.3	0.6	0.8	1.9	1.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.0	97.1	96.9	94.9	91.5	87.5	86.3	92.0	98.2	99.9	98.8	100.4	(19)
		MCWB	70.8	70.9	70.8	67.9	63.8	58.9	56.2	61.0	63.2	67.1	68.4	70.9	(20)
	2%	DB	92.0	92.7	93.2	90.2	86.2	83.1	82.3	88.0	93.3	95.1	95.1	94.9	(21)
		MCWB	72.0	71.2	73.2	68.2	61.9	58.1	57.1	60.1	62.9	66.5	68.1	71.6	(22)
	5%	DB	89.3	89.3	89.7	86.4	82.7	79.7	78.6	84.1	89.5	91.2	91.1	91.3	(23)
		MCWB	71.7	70.9	70.5	67.5	61.5	57.8	57.0	59.6	63.1	66.4	68.3	71.6	(24)
	10%	DB	86.1	86.2	86.1	82.7	79.2	76.6	75.2	80.0	84.9	87.0	87.2	87.8	(25)
		MCWB	71.2	70.6	70.3	66.9	61.2	57.7	56.8	58.7	62.6	66.0	68.5	70.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.9	77.4	77.1	74.2	68.5	63.9	63.4	66.4	69.1	73.2	75.7	77.5	(27)
		MCDB	86.4	83.0	89.0	86.9	80.9	76.9	73.7	80.1	85.0	86.4	86.0	85.9	(28)
	2%	WB	75.6	75.0	74.5	70.9	65.7	62.0	61.2	64.4	67.3	70.9	73.2	75.0	(29)
		MCDB	84.9	83.6	85.8	81.9	77.6	74.3	72.1	78.2	82.6	85.6	83.6	86.4	(30)
	5%	WB	74.3	73.4	73.3	69.3	64.3	60.7	59.8	62.8	65.7	69.2	71.8	73.6	(31)
		MCDB	83.6	82.3	83.3	80.4	76.5	72.5	70.8	75.1	80.8	83.3	82.3	85.1	(32)
	10%	WB	73.0	72.3	72.0	68.1	63.0	59.5	58.5	61.2	64.2	67.8	70.4	72.4	(33)
		MCDB	81.8	81.0	80.9	78.7	74.6	70.9	69.5	73.6	78.8	81.3	81.5	83.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	16.9	16.9	19.3	20.5	23.8	24.5	23.4	23.6	22.6	20.9	19.0	18.1	(35)
		MCDBR	22.1	22.2	25.6	27.2	30.0	29.9	29.6	30.9	31.0	28.7	27.5	24.6	(36)
	5% WB	MCWBR	6.5	6.1	8.2	8.9	11.1	12.1	11.9	11.2	10.6	8.6	8.2	7.5	(37)
		MCWBR	17.5	16.7	19.4	21.1	23.7	22.1	21.8	22.7	23.9	23.9	20.0	19.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.409	0.401	0.396	0.382	0.349	0.332	0.334	0.391	0.450	0.460	0.426	0.422	(40)
		taud	2.374	2.397	2.402	2.398	2.438	2.461	2.446	2.279	2.143	2.168	2.301	2.330	(41)
	Ebn at Noon		297	295	287	277	273	272	276	269	266	274	290	293	(42)
		Edh at Noon	42	40	38	35	30	29	30	39	48	49	44	43	(43)
(44) All-Sky Solar Radiation	RadAvg		1965	1915	1707	1457	1345	1204	1270	1481	1675	1748	1855	1932	(44)
	RadStd		134	150	124	96	58	72	45	89	106	110	156	161	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page